

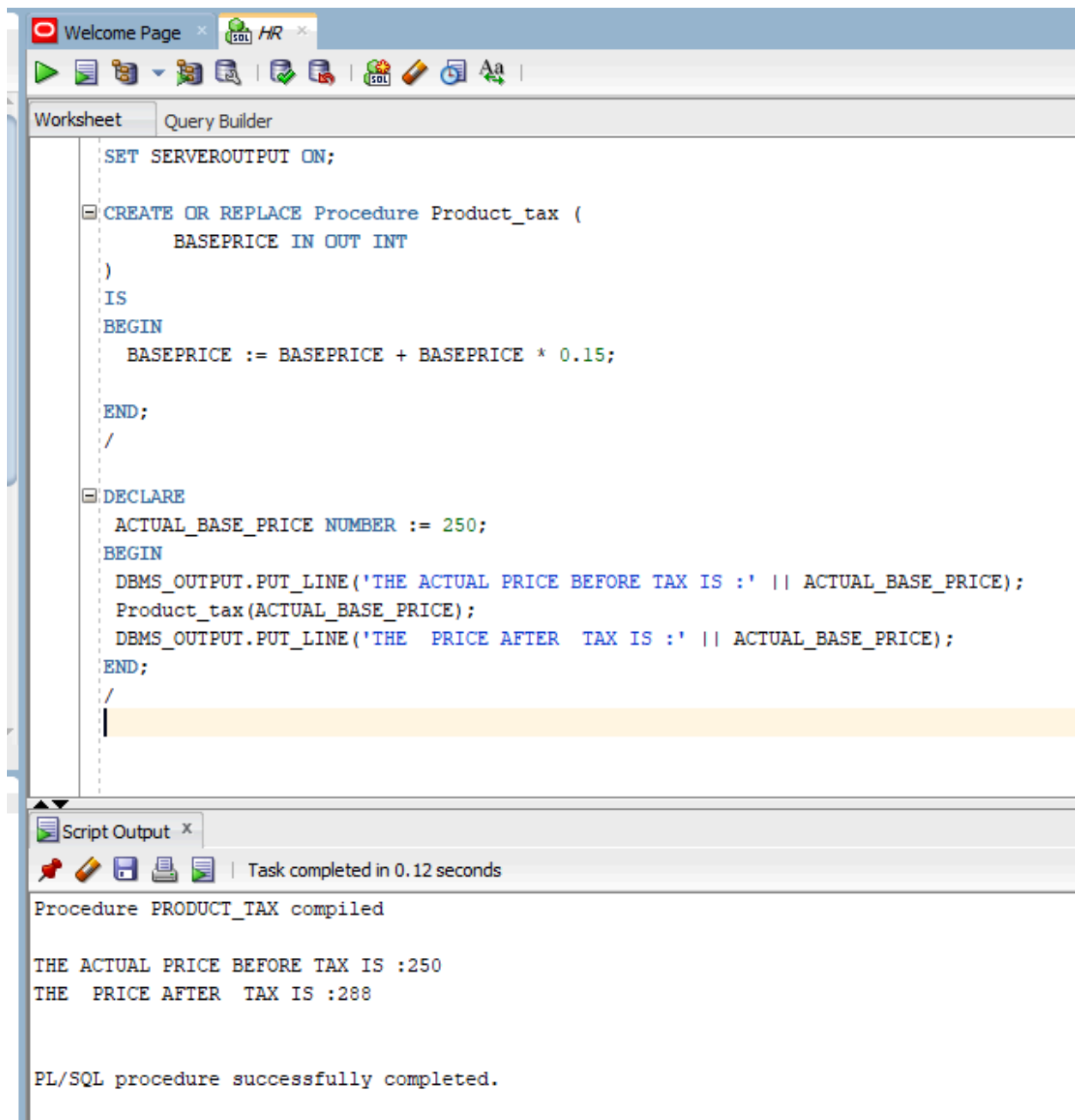
DBS LAB 10

24K-0559

BSCS 4H

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TASK#1).



The screenshot displays the Oracle SQL Developer environment. The main window is titled 'Query Builder' and contains a PL/SQL script. The script defines a procedure named 'Product\_tax' that takes a 'BASEPRICE' as input and returns an integer. The procedure calculates a 15% tax on the base price. Below the procedure definition, a 'DECLARE' block sets 'ACTUAL\_BASE\_PRICE' to 250 and calls the 'Product\_tax' procedure, printing the results using 'DBMS\_OUTPUT.PUT\_LINE'.

```
SET SERVEROUTPUT ON;

CREATE OR REPLACE Procedure Product_tax (
    BASEPRICE IN OUT INT
)
IS
BEGIN
    BASEPRICE := BASEPRICE + BASEPRICE * 0.15;

END;
/

DECLARE
    ACTUAL_BASE_PRICE NUMBER := 250;
BEGIN
    DBMS_OUTPUT.PUT_LINE('THE ACTUAL PRICE BEFORE TAX IS : ' || ACTUAL_BASE_PRICE);
    Product_tax(ACTUAL_BASE_PRICE);
    DBMS_OUTPUT.PUT_LINE('THE PRICE AFTER TAX IS : ' || ACTUAL_BASE_PRICE);
END;
/
```

The 'Script Output' window at the bottom shows the execution results. It indicates that the 'PRODUCT\_TAX' procedure was compiled successfully. The output shows the actual price before tax as 250 and the price after tax as 288. The final message states 'PL/SQL procedure successfully completed.'

Script Output x

Task completed in 0.12 seconds

Procedure PRODUCT\_TAX compiled

THE ACTUAL PRICE BEFORE TAX IS :250  
THE PRICE AFTER TAX IS :288

PL/SQL procedure successfully completed.

## TASK#2).

The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and editing. The 'Query Builder' tab is active, showing a PL/SQL script. The script sets server output on, selects all data from the HR.EMPLOYEES table, declares variables for employee first names and salaries, and then uses a loop to output the first name and salary for employee ID 105. The 'Script Output' window at the bottom shows the successful execution of the script, with the output: 'THE FIRST NAME IS :David and salary is 4800'.

```
SET SERVEROUTPUT ON;
select * from HR.EMPLOYEES;

DECLARE
  firstname HR.EMPLOYEES.FIRST_NAME%TYPE;
  monthyearn HR.EMPLOYEES.SALARY%TYPE;

BEGIN
  select e.FIRST_NAME,e.SALARY
  INTO firstname,monthyearn
  FROM HR.EMPLOYEES e
  where e.EMPLOYEE_ID = 105;

  DBMS_OUTPUT.PUT_LINE('THE FIRST NAME IS : ' || firstname || ' and salary is ' || monthyearn);

END;
/
```

Task completed in 0.056 seconds

PL/SQL procedure successfully completed.

THE FIRST NAME IS :David and salary is 4800

PL/SQL procedure successfully completed.

**TASK#3).**

The screenshot displays the Oracle SQL Developer environment. The top toolbar includes icons for running, saving, and editing. The 'Query Builder' tab is active, showing a PL/SQL procedure named 'monthlyearn' that updates the salary of employee 110 based on their current salary. The procedure uses a DECLARE statement for a variable, a SELECT statement to fetch the salary, an IF-THEN-ELSE block for conditional updates, and a COMMIT statement. The bottom pane shows the 'Script Output' window with the message 'PL/SQL procedure successfully completed.' and 'SALARY UPDATED'.

```
DECLARE
monthlyearn HR.EMPLOYEES.SALARY%TYPE;

BEGIN
select e.SALARY
INTO monthlyearn
FROM HR.EMPLOYEES e
where e.EMPLOYEE_ID = 110;

IF (monthlyearn < 8000) THEN
    UPDATE HR.EMPLOYEES SET SALARY = SALARY + 1200
    WHERE EMPLOYEE_ID = 110;
ELSE
    UPDATE HR.EMPLOYEES SET SALARY = SALARY + 400
    WHERE EMPLOYEE_ID = 110;

END IF;
COMMIT;
DBMS_OUTPUT.PUT_LINE('SALARY UPDATED ');

END;
/
```

Script Output x Query Result x

Task completed in 0.068 seconds

PL/SQL procedure successfully completed.

SALARY UPDATED

PL/SQL procedure successfully completed.

#### TASK#4).

```
DECLARE
    DEPARTMENTCODE HR.EMPLOYEES.DEPARTMENT_ID%TYPE;

BEGIN
    SELECT D.DEPARTMENT_ID
    INTO DEPARTMENTCODE
    FROM HR.EMPLOYEES D
    WHERE D.EMPLOYEE_ID = 115;

    IF (DEPARTMENTCODE = 90) THEN
        DBMS_OUTPUT.PUT_LINE('ADMINISTRATION.');
```

TECHNICAL SERVICES

```
    ELSE
        DBMS_OUTPUT.PUT_LINE('GENERAL STAFF.');
```

GENERAL STAFF.

```
    END IF;

END;
/
```

Script Output x Query Result x

Task completed in 0.052 seconds

PL/SQL procedure successfully completed.

GENERAL STAFF.

PL/SQL procedure successfully completed.

**TASK#5).**

Oracle SQL Developer interface showing a PL/SQL procedure execution.

**Query Builder**

```
DECLARE
    cost HR.EMPLOYEES.SALARY%TYPE;
    comm HR.EMPLOYEES.COMMISSION_PCT%TYPE;
    dept HR.EMPLOYEES.DEPARTMENT_ID%TYPE;
    bonus NUMBER;
BEGIN
    SELECT salary, commission_pct, department_id
    INTO cost, comm, dept
    FROM HR.EMPLOYEES
    WHERE employee_id = 100;

    IF dept = 90 THEN
        IF cost >= 15000 AND cost <= 20000 THEN
            bonus := cost * NVL(comm,0);
        ELSIF cost > 20000 THEN
            bonus := cost * NVL(comm,0) * 2;
        ELSE
            bonus := 0;
        END IF;

        DBMS_OUTPUT.PUT_LINE('Bonus: ' || bonus);
    ELSE
        DBMS_OUTPUT.PUT_LINE('Not Executive');
    END IF;
END;
```

**Script Output** | **Query Result**

Task completed in 0.092 seconds

PL/SQL procedure successfully completed.

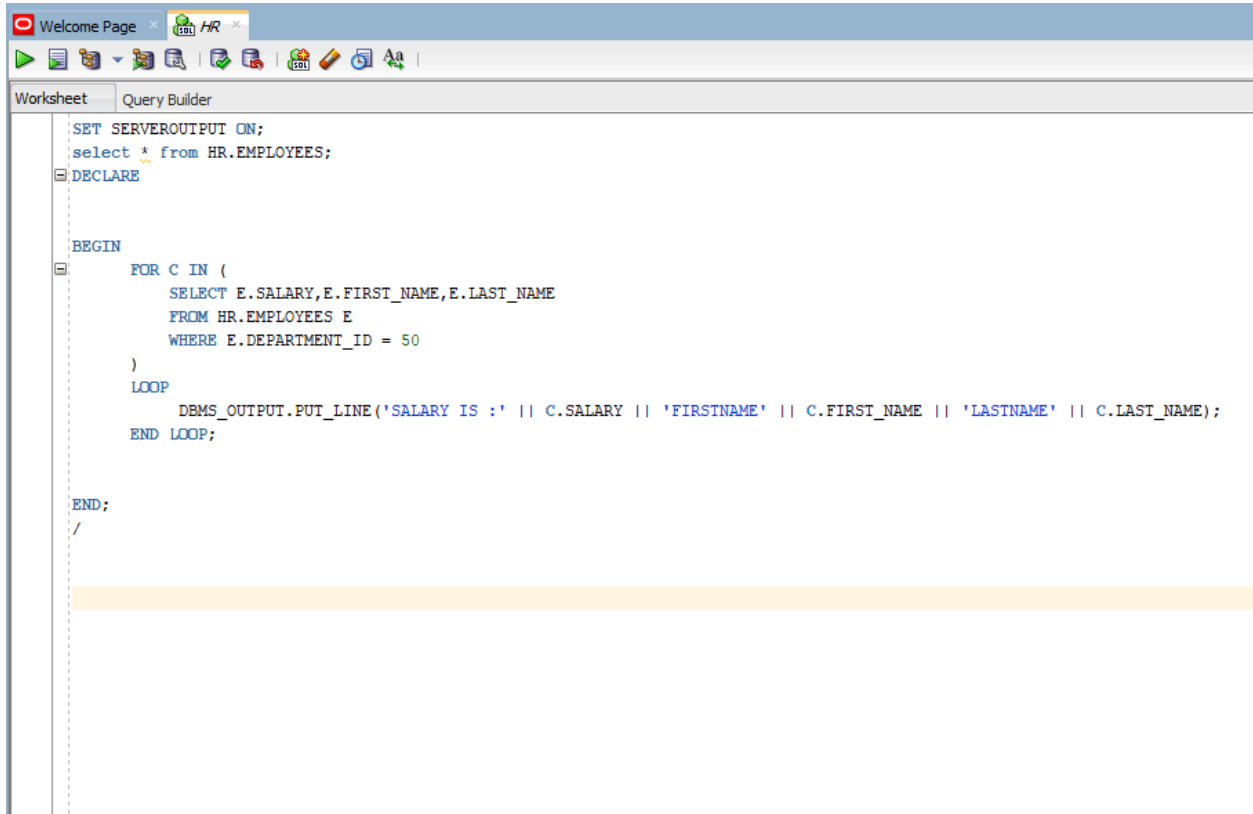
GENERAL STAFF.

PL/SQL procedure successfully completed.

Bonus: 0

PL/SQL procedure successfully completed.

## TASK#6).



The screenshot shows the Oracle SQL Developer interface. The top window is titled 'Welcome Page' and 'HR'. Below the toolbar, the 'Query Builder' tab is active. The main editor contains the following PL/SQL code:

```
SET SERVEROUTPUT ON;
select * from HR.EMPLOYEES;
DECLARE

BEGIN
  FOR C IN (
    SELECT E.SALARY, E.FIRST_NAME, E.LAST_NAME
    FROM HR.EMPLOYEES E
    WHERE E.DEPARTMENT_ID = 50
  )
  LOOP
    DBMS_OUTPUT.PUT_LINE('SALARY IS :' || C.SALARY || 'FIRSTNAME' || C.FIRST_NAME || 'LASTNAME' || C.LAST_NAME);
  END LOOP;

END;
/
```

The code is a PL/SQL block that sets server output on, selects all employees from the HR.EMPLOYEES table, and then loops through the results of a query that filters employees by department ID 50. For each employee, it prints their salary, first name, and last name concatenated together.

Welcome Page x HR x

Worksheet Query Builder

SET SERVEROUTPUT ON;

Script Output x Query Result x

Task completed in 0.053 seconds

```
SALARY IS :8000FIRSTNAMEMatthewLASTNAMEWeiss
SALARY IS :8200FIRSTNAMEAdamLASTNAMEFripp
SALARY IS :7900FIRSTNAMEPayamLASTNAMEKaufling
SALARY IS :6500FIRSTNAMEShantaLASTNAMEVollman
SALARY IS :5800FIRSTNAMEKevinLASTNAMEMourgos
SALARY IS :3200FIRSTNAMEJuliaLASTNAMENayer
SALARY IS :2700FIRSTNAMEIreneLASTNAMEMikkilineni
SALARY IS :2400FIRSTNAMEJamesLASTNAMELandry
SALARY IS :2200FIRSTNAMEStevenLASTNAMEMarkle
SALARY IS :3300FIRSTNAMELauraLASTNAMEBissot
SALARY IS :2800FIRSTNAMEMozheLASTNAMEAtkinson
SALARY IS :2500FIRSTNAMEJamesLASTNAMEMarlow
SALARY IS :2100FIRSTNAMETJLASTNAMEOlson
SALARY IS :3300FIRSTNAMEJasonLASTNAMEMallin
SALARY IS :2900FIRSTNAMEMichaelLASTNAMERogers
SALARY IS :2400FIRSTNAMEKiLASTNAMEGee
SALARY IS :2200FIRSTNAMEHazelLASTNAMEPhiltanker
SALARY IS :3600FIRSTNAMERenskeLASTNAMELadwig
SALARY IS :3200FIRSTNAMEStephenLASTNAMEStiles
SALARY IS :2700FIRSTNAMEJohnLASTNAMESeo
SALARY IS :2500FIRSTNAMEJoshuaLASTNAMEPatel
SALARY IS :3500FIRSTNAMETrennaLASTNAMERajs
SALARY IS :3100FIRSTNAMECurtisLASTNAMEDavies
SALARY IS :2600FIRSTNAMERandallLASTNAMEMatos
SALARY IS :2500FIRSTNAMEPeterLASTNAMEVargas
SALARY IS :3200FIRSTNAMEWinstonLASTNAMETaylor
SALARY IS :3100FIRSTNAMEJeanLASTNAMEFleaur
SALARY IS :2500FIRSTNAMEMarthaLASTNAMEsullivan
SALARY IS :2800FIRSTNAMEGirardLASTNAMEGeoni
SALARY IS :4200FIRSTNAMENanditaLASTNAMEsarchand
SALARY IS :4100FIRSTNAMEAlexisLASTNAMEBull
SALARY IS :3400FIRSTNAMEJuliaLASTNAMEDeLLinger
SALARY IS :3000FIRSTNAMEAnthonyLASTNAMECabrio
SALARY IS :3800FIRSTNAMEKellyLASTNAMEChung
SALARY IS :3600FIRSTNAMEJenniferLASTNAMEDilly
SALARY IS :2900FIRSTNAMETimothyLASTNAMEGates
SALARY IS :2500FIRSTNAMERandallLASTNAMEPerkins
SALARY IS :4000FIRSTNAMESarahLASTNAMEBell
SALARY IS :3900FIRSTNAMEBritneyLASTNAMEEverett
SALARY IS :3200FIRSTNAMESamuellLASTNAMEMcCain
SALARY IS :2800FIRSTNAMEVanceLASTNAMEJones
SALARY IS :3100FIRSTNAMEAlanaLASTNAMEWalsh
SALARY IS :3000FIRSTNAMEKevinLASTNAMEFeeney
SALARY IS :2600FIRSTNAMEDonaldLASTNAMEOConnell
```



Script Output x Query Result x

Task completed in 0.053 seconds

SALARY IS :2700FIRSTNAMEIreneLASTNAMEMikkilineni  
SALARY IS :2400FIRSTNAMEJamesLASTNAMELandry  
SALARY IS :2200FIRSTNAMEStevenLASTNAMEMarkle  
SALARY IS :3300FIRSTNAMELauraLASTNAMEBissot  
SALARY IS :2800FIRSTNAMEMozheLASTNAMEAtkinson  
SALARY IS :2500FIRSTNAMEJamesLASTNAMEMarlow  
SALARY IS :2100FIRSTNAMETJLASTNAMEOlson  
SALARY IS :3300FIRSTNAMEJasonLASTNAMEMallin  
SALARY IS :2900FIRSTNAMEMichaelLASTNAMERogers  
SALARY IS :2400FIRSTNAMEKiLASTNAMEGee  
SALARY IS :2200FIRSTNAMEHazelLASTNAMEPhiltanker  
SALARY IS :3600FIRSTNAMERenskeLASTNAMELadwig  
SALARY IS :3200FIRSTNAMEStephenLASTNAMEStiles  
SALARY IS :2700FIRSTNAMEJohnLASTNAMESeo  
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SALARY IS :3500FIRSTNAMETrennaLASTNAMERajs  
SALARY IS :3100FIRSTNAMECurtisLASTNAMEDavies  
SALARY IS :2600FIRSTNAMERandallLASTNAMEMatos  
SALARY IS :2500FIRSTNAMEPeterLASTNAMEVargas  
SALARY IS :3200FIRSTNAMEWinstonLASTNAMETaylor  
SALARY IS :3100FIRSTNAMEJeanLASTNAMEFleaur  
SALARY IS :2500FIRSTNAMEMarthaLASTNAMESullivan  
SALARY IS :2800FIRSTNAMEGirardLASTNAMEGeoni  
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SALARY IS :3400FIRSTNAMEJuliaLASTNAMEDellinger  
SALARY IS :3000FIRSTNAMEAnthonyLASTNAMECabrio  
SALARY IS :3800FIRSTNAMEKellyLASTNAMEChung  
SALARY IS :3600FIRSTNAMEJenniferLASTNAMEDilly  
SALARY IS :2900FIRSTNAMETimothyLASTNAMEGates  
SALARY IS :2500FIRSTNAMERandallLASTNAMEPerkins  
SALARY IS :4000FIRSTNAMESarahLASTNAMEBell  
SALARY IS :3900FIRSTNAMEBritneyLASTNAMEEverett  
SALARY IS :3200FIRSTNAMESamuellLASTNAMEMcCain  
SALARY IS :2800FIRSTNAMEVanceLASTNAMEJones  
SALARY IS :3100FIRSTNAMEAlanaLASTNAMEWalsh  
SALARY IS :3000FIRSTNAMEKevinLASTNAMEFeeney  
SALARY IS :2600FIRSTNAMEDonaldLASTNAMEOConnell  
SALARY IS :2600FIRSTNAMEDouglasLASTNAMEGrant

PL/SQL procedure successfully completed.

**TASK#7)**

Oracle SQL Developer interface showing a PL/SQL procedure and its execution results.

**Query Builder**

```
SET SERVEROUTPUT ON;
select * from HR.;
CREATE OR REPLACE PROCEDURE get_salary (
    p_id IN NUMBER
)
IS
    v_salary HR.EMPLOYEES.SALARY%TYPE;
BEGIN
    SELECT salary INTO v_salary
    FROM HR.EMPLOYEES
    WHERE employee_id = p_id;

    DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
END;
/

BEGIN
    get_salary(100);
END;
/
```

**Script Output** | **Query Result**

Task completed in 0.141seconds

```
SALARY IS :2600FIRSTNAMEDonaldLASTNAMEOConnell
SALARY IS :2600FIRSTNAMEDouglasLASTNAMEGrant

PL/SQL procedure successfully completed.

Procedure GET_SALARY compiled

Salary: 24000

PL/SQL procedure successfully completed.
```

**TASK#8)**

Oracle SQL Developer interface showing a PL/SQL script in the Query Builder.

Worksheet | Query Builder

```
SET SERVEROUTPUT ON;
select * from HR.;
CREATE OR REPLACE PROCEDURE get_dept (
    p_id IN NUMBER,
    p_dept OUT NUMBER
)
IS
BEGIN
    SELECT department_id INTO p_dept
    FROM HR.EMPLOYEES
    WHERE employee_id = p_id;
END;
/

DECLARE
    v_dept NUMBER;
BEGIN
    get_dept(100, v_dept);
    DBMS_OUTPUT.PUT_LINE('Department: ' || v_dept);
END;
/
```

Script Output x | Query Result x

Task completed in 0.109 seconds

Salary: 24000

PL/SQL procedure successfully completed.

Procedure GET\_DEPT compiled

Department: 90

PL/SQL procedure successfully completed.

**TASK#9)**

Welcome Page x SQL HR x



WorksheetQuery Builder

```
SET SERVEROUTPUT ON;
select * from HR.;
CREATE OR REPLACE PROCEDURE increase_val (
    p_val IN OUT NUMBER
)
IS
BEGIN
    p_val := p_val + (p_val * 0.25);
END;
/

DECLARE
    v_num NUMBER := 200;
BEGIN
    increase_val(v_num);
    DBMS_OUTPUT.PUT_LINE('Updated Value: ' || v_num);
END;
/
```

Script Output xQuery Result x



Task completed in 0.163 seconds

Department: 90

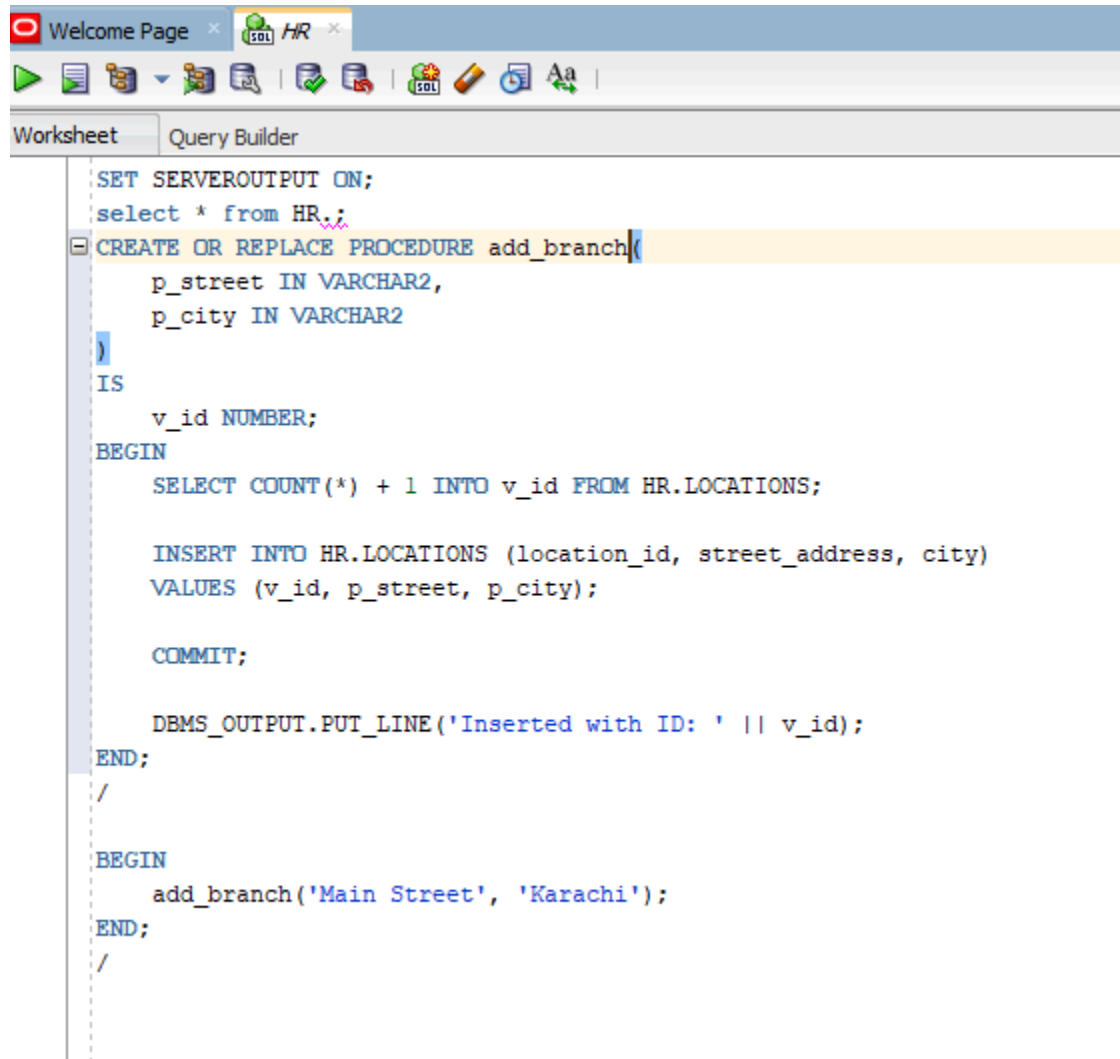
PL/SQL procedure successfully completed.

Procedure INCREASE\_VAL compiled

Updated Value: 250

PL/SQL procedure successfully completed.

## TASK#10)



The screenshot shows a SQL query editor window with a toolbar at the top containing icons for running queries, saving, and other functions. The window has two tabs: 'Welcome Page' and 'HR'. Below the tabs is a toolbar with icons for undo, redo, and other editing functions. The main area of the window is divided into two sections: 'Worksheet' and 'Query Builder'. The 'Query Builder' section is active, displaying a PL/SQL procedure named 'add\_branch'.

```
SET SERVEROUTPUT ON;
select * from HR.;
CREATE OR REPLACE PROCEDURE add_branch(
    p_street IN VARCHAR2,
    p_city IN VARCHAR2
)
IS
    v_id NUMBER;
BEGIN
    SELECT COUNT(*) + 1 INTO v_id FROM HR.LOCATIONS;

    INSERT INTO HR.LOCATIONS (location_id, street_address, city)
    VALUES (v_id, p_street, p_city);

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Inserted with ID: ' || v_id);
END;
/

BEGIN
    add_branch('Main Street', 'Karachi');
END;
/
```